Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	29 October 2026
Team ID	N/A - individual project; no Team ID provided
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Read the cleaned CSV for Tableau and use the SQLite table ev_market_clean for SQL/API operations.	High
2	Data Processing	Remove one exact duplicate; median-impute missing numeric values in five columns; preserve categorical values; document synthetic source.	High
3	Business Analysis	Calculate record count, total recorded sales, average range/price/rating, and group by category, manufacturer, region, model year, country and battery type.	High
4	Business Insights & Outputs	Provide nine distinct Tableau visualizations, an interactive dashboard, three-scene Story and a Flask web page.	High
5	User Interaction & Analysis	Offer Market Region, Category, Model Year and Battery Type filters; support tooltips and view navigation.	High
6	Data Update & Reporting	Use a reproducible data-preparation/SQLite setup and document that the current analysis is a static dataset extract; future updates require refreshed data and republishes.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Readable chart titles, clear axes, consistent visual style, four filters, usable tooltips and clearly titled Story scenes.
NFR-2	Security	No secrets in the public repo; .env excluded; note Tableau Public data/views are public; synthetic dataset only.
NFR-3	Reliability	SQLite contains 199,999 records and 199,999 distinct vehicle IDs; local Flask endpoint tests were reported as passing.
NFR-4	Performance	Use an extracted CSV in Tableau Public; timings were not instrumented, so response/render latency remains unmeasured.
NFR-5	Availability	Flask web app deployed to Render at the provided URL; Tableau embeds depend on public Tableau view availability.
NFR-6	Scalability	Current student prototype uses a local SQLite DB and a static dataset; larger workloads would need managed storage, caching and access controls.